

Sparse-View CT Reconstruction on Mayo Data

Variational and Learning-Based Methods for Computational Imaging

Group B – Computational Imaging Project

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Abstract

This report studies sparse-view computed tomography reconstruction as an inverse problem. The experimental pipeline uses Mayo CT slices resized to 256×256 , simulates parallel-beam measurements with 180, 90, 60, and 45 projection angles, and adds relative Gaussian measurement noise with level 0.005. The same degraded sinograms are shared by all reconstruction methods. We compare a model-based Total p -Variation method solved with a Chambolle–Pock primal-dual algorithm and a learned image-domain reconstruction method based on a generalized IPPy U-Net/ResUNet post-processor. The final report will quantify performance with PSNR and SSIM and discuss visual artifacts, stability across view counts, and the trade-off between model-based regularization and supervised learning.

1 Introduction

Sparse-view CT reduces the number of projection angles acquired by the scanner, lowering acquisition time and potentially radiation exposure, but it makes the inverse problem more ill-posed. Filtered backprojection can produce severe streak artifacts when the angular sampling is limited. The aim of this project is therefore to compare reconstruction strategies from different methodological families under the same acquisition protocol and on the same test data.

The project specifications require:

- Mayo CT data, resized to 256×256 ;
- sparse-view parallel-beam CT with $n_\theta \in \{180, 90, 60, 45\}$;
- additive measurement noise with relative level 0.005;
- a shared degraded input for all tested methods;
- quantitative comparison through PSNR and SSIM, plus visual comparison panels.

The report follows the style of a short computational imaging paper: after defining the inverse problem, it describes the data pipeline, reconstruction methods, evaluation protocol, results, and limitations.

2 Problem Formulation

Let $x \in \mathbb{R}^{256 \times 256}$ denote a clean CT slice and let K_θ be the discrete parallel-beam CT projection operator for a given number of views θ . The observed sinogram is modeled as

$$y^\delta = K_\theta x + e, \tag{1}$$

where e is additive Gaussian measurement noise scaled so that

$$\frac{\|e\|_2}{\|K_\theta x\|_2} \approx 0.005. \tag{2}$$

Table 1: Dataset split summary. Fill the counts from `processed2/manifest.json`.

Split	Number of patients	Number of slices
Train	TODO: fill	TODO: fill
Validation	TODO: fill	TODO: fill
Test	TODO: fill	TODO: fill

The reconstruction task is to estimate x from y^δ . Since angular undersampling makes the inverse problem ill-posed, prior information is required either explicitly, through regularization, or implicitly, through a learned reconstruction model.

3 Dataset and Preprocessing

The dataset is the Mayo CT dataset specified for Group B. The preprocessing stage is implemented in notebook `00_data_and_degradation.ipynb`. Each slice is loaded as a grayscale image, resized to 256×256 , normalized to $[0, 1]$, and stored in patient-level PyTorch files. For each patient and each split, the pipeline stores:

- the clean image stack;
- noisy sinograms for 180, 90, 60, and 45 views;
- metadata including patient identifier, geometry, detector size, and empirical relative noise.

The data contract is important for fairness: all downstream methods read the same saved sinograms rather than regenerating degraded observations independently. This avoids hidden differences in noise realizations or projection geometry.

4 Methods

4.1 Filtered Backprojection Proxy

Filtered backprojection (FBP) is not treated as a final project method in the current comparison, but it is used as the degraded image-domain input to the neural network. FBP is computed with the IPPy wrapper `solvers.FBP`, which delegates to the ASTRA-based `CTProjector.FBP`. This gives a consistent image-domain proxy for all view counts.

4.2 Total p -Variation Reconstruction

The variational method solves

$$\hat{x} = \arg \min_{x \geq 0} \frac{1}{2} \|K_\theta x - y^\delta\|_2^2 + \lambda R_p(x), \quad (3)$$

where R_p is a Total p -Variation prior and $p \in (0.1, 0.5)$ as required by the project specifications. In the current implementation, $p = 0.35$ and $\lambda = 0.01$. The optimization is performed with IPPy `solvers.ChambollePockTpVUnconstrained`, a primal-dual scheme inspired by Chambolle–Pock algorithms [1]. The reported PSNR and SSIM are those computed internally by the solver diagnostics, so the metrics refer to the raw optimization iterate rather than to a display-normalized image.

Table 2: Main experimental parameters.

Parameter	Value
Image size	256×256
Geometry	Parallel beam
Detector size	256
Views	180, 90, 60, 45
Relative noise level	0.005
TpV p	0.35
TpV λ	0.01
TpV iterations	150
ResUNet epochs	TODO: fill final value
ResUNet loss	Mean squared error

4.3 Generalized ResUNet Post-Processing

The end-to-end learning method is implemented in notebook `02_ResUnet_reconstruction.ipynb`. A single generalized IPPy U-Net is trained across all four view configurations. The input is the FBP reconstruction of the noisy sparse-view sinogram; the target is the clean CT image. The model uses IPPy residual down/up blocks and a final sigmoid activation, so its output is naturally constrained to $[0, 1]$.

This choice is motivated by the U-Net architecture [2]: encoder-decoder skip connections preserve spatial detail, while the bottleneck captures larger context. Training one model across all view counts encourages robustness to different artifact strengths. The current checkpoint strategy saves a latest checkpoint at the end of each epoch and resumes from the next epoch when requested.

4.4 Additional Method Slot

The project specifications for a group of two students require three of the four methodological families. The current codebase contains the variational and end-to-end methods. If the final project includes a third method, this section should describe it. Two compatible options are:

- a hybrid learned primal-dual method, following the unrolled primal-dual idea of Adler and Öktem [3], potentially using `IPPy.hybrid.LearnedPrimalDual`;
- a diffusion-based method, using a reconstruction prior and explaining how the data consistency step is adapted to sparse-view CT.

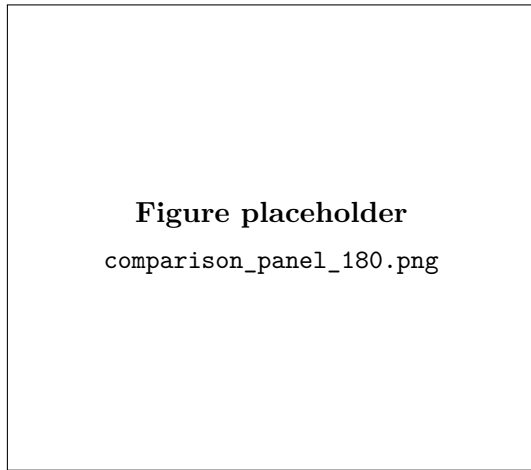
If no third method is implemented, the report should explicitly justify the final methodological scope during the oral discussion.

5 Experimental Setup

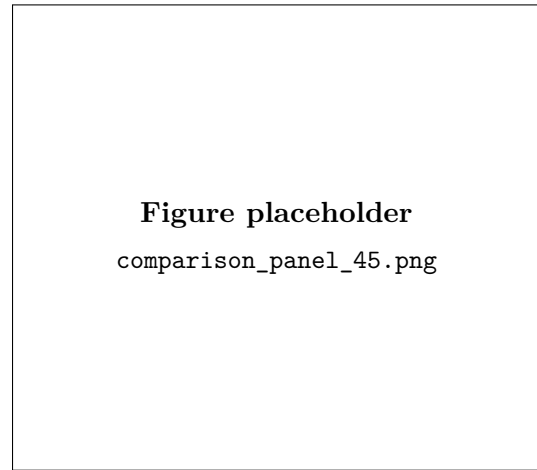
All methods are evaluated on the same central test slice and the same saved sinograms. This is the main fairness constraint: each method receives identical degraded data for a fixed view count. Quantitative quality is measured with PSNR and SSIM [4]. Visual quality is assessed through reconstruction panels and absolute error maps.

Table 3: PSNR and SSIM comparison on the shared test slice.

Views	Method	PSNR (dB)	SSIM
180	TpV	TODO: fill	TODO: fill
180	ResUNet	TODO: fill	TODO: fill
90	TpV	TODO: fill	TODO: fill
90	ResUNet	TODO: fill	TODO: fill
60	TpV	TODO: fill	TODO: fill
60	ResUNet	TODO: fill	TODO: fill
45	TpV	TODO: fill	TODO: fill
45	ResUNet	TODO: fill	TODO: fill



(a) 180 views



(b) 45 views

Figure 1: Example comparison panels. If the output figures are not generated yet, placeholders are shown.

6 Results

6.1 Quantitative Comparison

Table 3 should be filled with the values printed by `04_results_comparison.ipynb`. TpV metrics must be read from the IPPy solver diagnostics, while ResUNet metrics must be computed on the final clamped network output, without per-image renormalization.

6.2 Visual Comparison

Figure 1 should include representative visual panels generated by the final comparison notebook. The expected qualitative trend is that fewer views produce stronger streak artifacts and a harder inverse problem. The discussion should compare edge preservation, noise suppression, anatomical detail, and possible oversmoothing.

7 Discussion

The discussion should connect quantitative results to visible artifacts. A possible structure is:

- **Effect of sparse views.** Explain how performance changes from 180 to 45 views and whether each method degrades smoothly.

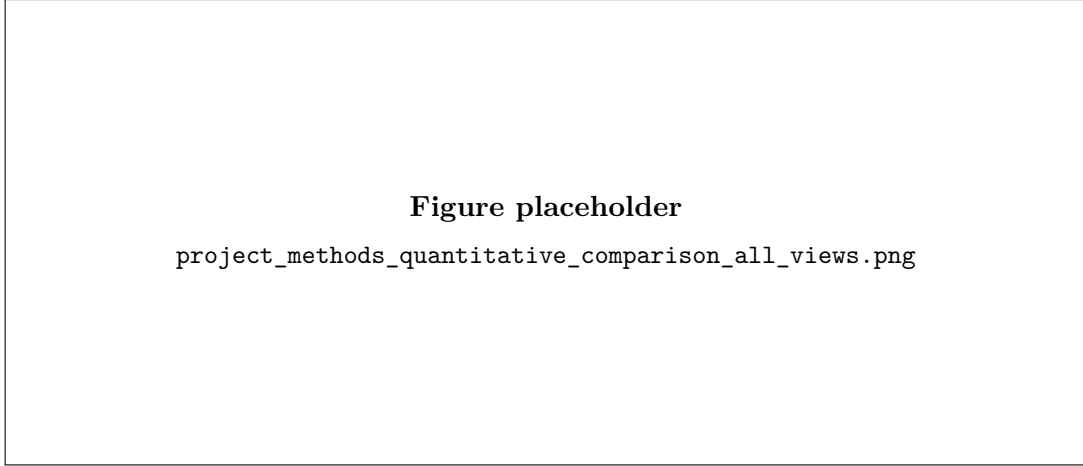


Figure 2: Aggregate PSNR/SSIM plot across view counts.

- **TpV behavior.** Discuss regularization strength, edge preservation, staircasing or over-smoothing, and convergence curves.
- **ResUNet behavior.** Discuss artifact suppression, possible hallucination/oversmoothing, and dependence on FBP input quality.
- **Fairness.** Emphasize that all methods use the same stored sinograms and the same test slice.
- **Limitations.** Mention single-slice reporting if only a representative slice is used, limited hyperparameter sweep, and the need for an additional method if required by final evaluation rules.

8 Conclusions

This project frames sparse-view CT as an inverse problem and compares model-based and learning-based reconstruction strategies under a shared degradation protocol. TpV offers an interpretable regularized solution with explicit data fidelity and prior terms. The generalized ResUNet provides a supervised artifact-reduction model trained on FBP proxies across all view counts. The final conclusion should be written after inserting the measured PSNR/SSIM values and inspecting the generated visual panels.

References

- [1] Antonin Chambolle and Thomas Pock. A first-order primal-dual algorithm for convex problems with applications to imaging. *Journal of Mathematical Imaging and Vision*, 40(1):120–145, 2011.
- [2] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-net: Convolutional networks for biomedical image segmentation. In *Medical Image Computing and Computer-Assisted Intervention*, pages 234–241, 2015.
- [3] Jonas Adler and Ozan Öktem. Learned primal-dual reconstruction. *IEEE Transactions on Medical Imaging*, 37(6):1322–1332, 2018.
- [4] Zhou Wang, Alan C. Bovik, Hamid R. Sheikh, and Eero P. Simoncelli. Image quality assessment: From error visibility to structural similarity. *IEEE Transactions on Image Processing*, 13(4):600–612, 2004.